Chris's spreadsheet model: column definitions and comparison with $A_F(\ell)$

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1 Workbook

File: `b3737b04....xlsx` (copied to `chris_model.xlsx`). The per-angle model is on the tab *density w rotation* and its three copies; all four tabs have identical row formulas and differ only in parameter cells (Table 1). Numbers below are from *Copy of Copy of Copy of density*, the tab recorded as the working set in our 18 Sep notes. Rows 66–426 hold $\theta = 0, 1, \ldots, 360°$. The final column is **BH**, "Model A/A_all"; the sheet's chart "a and migkas" plots BH against the Migkas column BI.

The other two workbooks are not this model: `9033....xlsx` is our own calculator (`A_vs_angle_calc.xlsx`), and `37301....xlsx` is an earlier point-mass version (the *point* tab here; no A column, force $\propto 1/r$).

Geometry, as the formulas define it: plane (x, y), centre of expansion (CoE) at the origin, Earth at $(D, 0)$. The source P lies on a circle of radius R about Earth, at viewing angle θ measured from $+x$; $\theta = 0$ looks directly away from the CoE and $\theta = 180°$ looks at it.

2 Parameter cells

Column letters in sans serif (C, L, ...) are sheet columns in the same row; italic symbols are parameter cells.

Cell	Label (his)	Formula	Value
C23	–	input	5 $(= D/R)$
D23	–	input	0.18 $(= L/D)$
B25	k "1/s, Taylor Sedov Hubble constant"	input	2.18×10^{-18}
B26	G "N(km/kg)^2"	input	6.67×10^{-17}
B27	M "kg"	$= \mathrm{D27} \cdot \mathrm{E27} \cdot 10^{13}$, D27 $= 10^{53}$, E27 $= 444$	4.44×10^{68}
B28	D "km, Earth to coe"	$= \mathrm{B29} \cdot \mathrm{C23}$	6.5×10^{23}
B29	R "km, Earth to P"	input	1.3×10^{23}
B30	L "km, density distribution"	$= \mathrm{B28} \cdot \mathrm{D23}$	1.17×10^{23}
B32	V_o "observer velocity, all x"	$= kD$	1.417×10^{6}
B33	dt	input	1
B34	$A_{x,\mathrm{obs}}$ "observer acceleration, all x"	$= -GM/D^2$	-7.009×10^{4}
B36	v^*	$= Rk$	2.834×10^{5}
B55	f_D "fraction up to o"	$= 1 - e^{-D/L}(1 + D/L + D^2/2L^2)$	0.91500
B56	M_{tot} "total mass"	$= M/f_D$	4.852×10^{68}
E39	–	input (subtracted from Migkas B)	0.09
BA57	amplitude, A rotation	input	-0.06
BA58	ω	$= 2\pi/360$	0.0174533
BA59	phase ϕ (rad)	input	0.35
BE57–59	B rotation	$= -$BA57, BA58, BA59	$+0.06$, ω, 0.35

Cells not used by any row formula: B7–B10 ($c = 100$, v/c, $\sqrt{1 - v^2/c^2}$, which shows `#NUM!`), unlabelled B12–B18, B23 ($L = 0.5$), D27–E28, D29 $= (D/R)^3$, B54, C54, D54.

3 Columns, in sheet order

Dependencies are listed after each header. Row 66 differs from the other rows only in column G (noted there).

A "degrees". Viewing angle θ from Earth. Input.

$$\mathsf{A} = \theta \tag{1}$$

B "radians". From A.

$$\mathsf{B} = \pi\,\mathsf{A}/180 \tag{2}$$

C "px". x of P. From B; D, R.

$$\mathsf{C} = D + R\cos\mathsf{B} \tag{3}$$

D "py". y of P. From B; R.

$$\mathsf{D} = R \sin \mathsf{B} \tag{4}$$

E "D". Distance of P from the CoE. From C, D.

$$\mathsf{E} = \sqrt{\mathsf{C}^2 + \mathsf{D}^2} \tag{5}$$

F "V=kD". Sedov–Taylor speed of P about the CoE. From E; k.

$$\mathsf{F} = k\,\mathsf{E} \tag{6}$$

G "theta". Polar angle of P seen from the CoE. From C, D. Row 66 adds `RADIANS(B58)`; B58 is empty, so it adds 0.

$$\mathsf{G} = \arctan(\mathsf{D}/\mathsf{C}) \tag{7}$$

H "Vx". From F, G.

$$\mathsf{H} = \mathsf{F} \cos \mathsf{G} \tag{8}$$

I "Vy". From F, G.

$$\mathsf{I} = \mathsf{F} \sin \mathsf{G} \tag{9}$$

J "dVx". Velocity of P relative to Earth, x. From H; V_o.

$$\mathsf{J} = \mathsf{H} - V_o \tag{10}$$

K "dVy". Same, y (Earth has no y velocity). From I.

$$\mathsf{K} = \mathsf{I} \tag{11}$$

L "recession v*". Relative speed without gravity. From J, K. Algebraically $\mathsf{J} = kR\cos\theta$ and $\mathsf{K} = kR\sin\theta$, so $\mathsf{L} = kR = v^*$ in every row.

$$\mathsf{L} = \sqrt{\mathsf{J}^2 + \mathsf{K}^2} \tag{12}$$

AO "mass up to P". Mass inside radius E for $\rho \propto e^{-r/L}$, scaled so that the mass inside D equals M. From E; L, M_{tot}. (It sits in column AO, but N uses it.)

$$\mathsf{AO} = \left[1 - e^{-\mathsf{E}/L}\left(1 + \frac{\mathsf{E}}{L} + \frac{\mathsf{E}^2}{2L^2}\right)\right] M_{\mathrm{tot}} \tag{13}$$

N "A". Gravitational acceleration of P toward the CoE. From E, AO; G.

$$\mathsf{N} = -G\,\mathsf{AO}/\mathsf{E}^2 \tag{14}$$

O "Ax". From N, G.

$$\mathsf{O} = \mathsf{N} \cos \mathsf{G} \tag{15}$$

P "Ay". From N, G.

$$\mathsf{P} = \mathsf{N} \sin \mathsf{G} \tag{16}$$

Q "dAx". Acceleration of P relative to Earth, x. From O; $A_{x,\mathrm{obs}}$.

$$\mathsf{Q} = \mathsf{O} - A_{x,\mathrm{obs}} \tag{17}$$

R "dAy". Same, y. From P.

$$\mathsf{R} = \mathsf{P} \tag{18}$$

S "dVx*". Relative velocity after one step of relative acceleration, x. From J, Q; dt.

$$\mathsf{S} = \mathsf{J} + \mathsf{Q}\,dt \tag{19}$$

T "dVy*". Same, y. From K, R; dt. The terms are in the opposite order to S (see item 1 in Sec. 6).

$$\mathsf{T} = \mathsf{R} + \mathsf{K}\,dt \tag{20}$$

U "recession v". Relative speed with gravity. From S, T.

$$\mathsf{U} = \sqrt{\mathsf{S}^2 + \mathsf{T}^2} \tag{21}$$

X "angle v". Direction of (S, T); a diagnostic, not used downstream. Excel `ATAN2(S,T)`.

$$\mathsf{X} = \operatorname{atan2}(\mathsf{T}, \mathsf{S}) \tag{22}$$

Y "angle a". Direction of the acceleration of P; a diagnostic. From O, P.

$$\mathsf{Y} = \arctan(\mathsf{P}/\mathsf{O}) \tag{23}$$

AB, AC "recession v*, (recession v)". Plot copies. From L, U.

$$\mathsf{AB} = \mathsf{L}, \qquad \mathsf{AC} = \mathsf{U} \tag{24}$$

AF "(recession v); plotted as "A/A_all Model"". Ratio of speed with gravity to speed without gravity. From U, AB.

$$\mathsf{AF} = \mathsf{U}/\mathsf{AB} \tag{25}$$

BA "model". Sine "rotation" term in the sheet angle θ; $\omega\theta$ is in radians. From A (through AR = AJ = AA = A); BA57–59.

$$\mathsf{BA} = \mathrm{BA57}\,\sin(\omega\,\theta + \phi) \tag{26}$$

BH "Model A/A_all". **Final A column.** From AF, BA.

$$\mathsf{BH} = (1 - \mathsf{BA})\,\mathsf{AF} \tag{27}$$

The B branch: **AN** "mass up to p x" = AO/M; **BE** = $\mathrm{BE57}\sin(\omega\theta + \phi)$; **BL** "Model B/B_all" = $(1 - \mathsf{BE})\,\mathsf{AN}$. Written out, with dt kept:

$$\mathsf{BH}(\theta) = \big[1 - a_A \sin(\omega\theta + \phi)\big]\,\frac{\big|\,kR\,\hat{\mathbf{n}}(\theta) + dt\,[\mathbf{g}(\mathbf{r}_P) - \mathbf{g}(D\hat{\mathbf{x}})]\,\big|}{kR}, \qquad \mathbf{g}(\mathbf{r}) = -\frac{G\,m(r)}{r^2}\hat{\mathbf{r}}, \tag{28}$$

where $a_A = \mathrm{BA57}$ and $m(r) = \mathsf{AO}$. The y part matches this only because $dt = 1$ (item 1).

Comparison columns (no effect on BH). AD, AL: angle, copied. AG = `VLOOKUP(AD`$_{n+10}$`, migkas!E:H, 3, TRUE)` gives Migkas A; AM is the same lookup for Migkas B, minus E39. AE = AG; BI = AG; BM = AM. AH = |AF − AG|, AI = (AF − AG)/AF, AP = |AM − AN|, AQ = (AN − AM)/AN; AZ = AI and BD = AQ. BJ = (BH − BI)/BH, BN = (BL − BM)/BL. AK = $10^{\mathsf{AF}}/2$. AS = AF shifted by 180°. AV–AY are helper columns. Summary cells: AF49, AM49, BH56 and BL56 are `LINEST` fits; BH58 is the R^2 of BI on BH (0.7389); D3 = BH58/AF51 "R2w/R2wo".

4 Recomputation check

`recompute_columns.py` implements Eqs. (1)–(27) literally and compares them with the values cached in the file. Model columns B–BL agree to a maximum relative difference of 4×10^{-6} (primary tab; $\le 2 \times 10^{-5}$ on all four tabs). The residual differences come from the file storing about 7 significant figures. For BH alone the difference is 1.5×10^{-7}. The lookup columns AG and AM agree except on 7 and 8 rows. On those rows $\theta + 10$ is a multiple of 8, so the lookup key falls exactly on a node of the 360/315 grid and the sheet's rounding decides which side is taken. Using the cached AG, the sheet's own R^2 is reproduced (0.73894 vs BH58 = 0.73894).

5 Comparison with $A_F(\ell)$ and the Migkas points

$A_F(\ell) = 0.9965 + 0.0296\cos\ell + 0.0434\sin\ell + 0.0636\cos 2\ell + 0.0366\sin 2\ell - 0.0299\cos 3\ell - 0.0114\sin 3\ell$. Two angle mappings are used. (i) $\theta = \ell$. (ii) The sheet's lookup mapping, which pairs row θ with Migkas $\ell = 45 + \lfloor \frac{7}{8}(\theta + 10) \rfloor$ (item 4). Here $R^2 = 1 - \sum(y - \mathsf{BH})^2 / \sum(y - \bar{y})^2$ has no refit; r^2 is the squared correlation, which is what `LINEST` reports.

Comparison (primary tab)	N	RMS	R^2	r^2
BH vs A_F, $\theta = \ell$, 0–359°	360	0.0376	0.691	0.779
AF (no sine term) vs A_F, $\theta = \ell$	360	0.0522	0.406	–
BH vs 71 Migkas points, $\theta = \ell$	71	0.0429	0.632	0.712
BH vs Migkas points, sheet mapping (ii)	62	0.0576	0.259	0.532
A_F vs 71 Migkas points	71	0.0206	0.915	–
Sheet's own BI vs BH (cell BH58)	361	–	–	0.739

$\theta = \ell$ (°)	0	30	60	90	120	150	180	210	240	270	300	330
BH $- A_F$	.029	−.002	−.024	.019	.082	.080	.016	−.022	−.001	.021	.017	.023

The largest residual is +0.092 at 134°. There A_F and the data dip to ≈ 0.89 near 125°, while BH stays near 1.0. The mean of BH over θ is 1.0164, against 0.9965 for A_F. Elsewhere BH follows A_F to within about 0.03: the maximum near 20°, the second maximum near 180°, and the minimum near 260°.

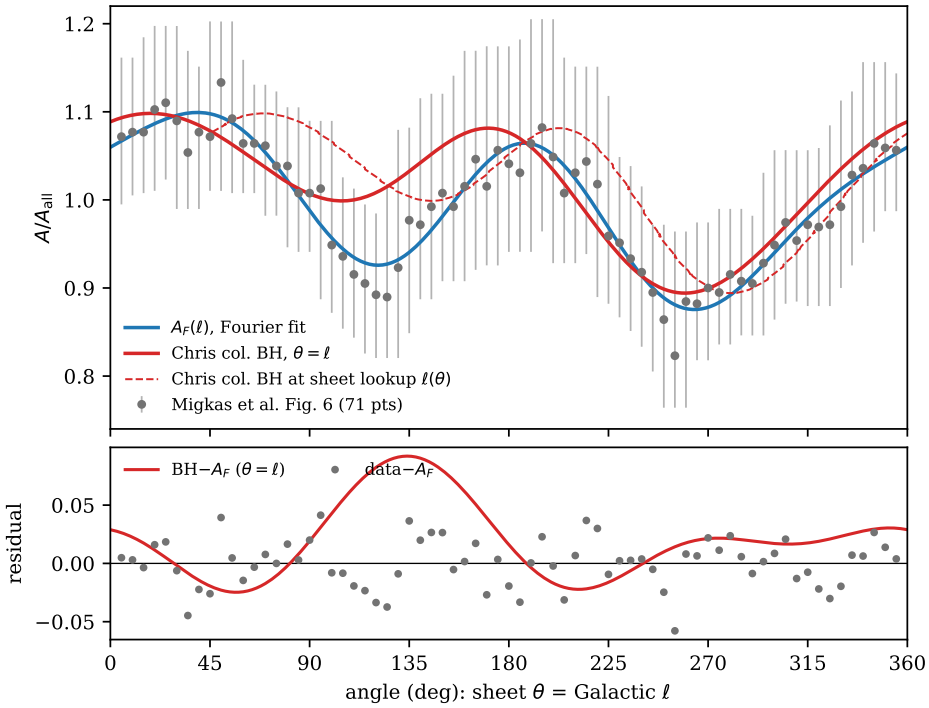


Figure 1: Top: BH (primary tab) at $\theta = \ell$ (solid) and at the sheet's lookup mapping (dashed), A_F, and the Migkas points. Bottom: BH$-A_F$ and data$-A_F$.

Tab	D/R	L/D	$GM\,dt/D^2kR$	BA57	ϕ	BH58	RMS vs A_F	R^2 vs data
density w rotation	3.33	0.150	0.167	−0.055	0.15	0.686	0.0436	0.531
Copy of density w rot.	9.00	0.250	0.414	−0.06	0	0.594	0.0416	0.566
Copy of Copy of ...	6.10	0.199	0.331	−0.06	0.40	0.745	0.0369	0.644
Copy of Copy of Copy (primary)	5.00	0.180	0.247	−0.06	0.35	0.739	0.0376	0.632

Table 1: The four working tabs (R^2 vs data: 71 points, $\theta = \ell$). The fifth copy replaces S and V_o with escape-speed expressions; its cached BH58 is 8×10^{-4} and D3 shows `#DIV/0!`, so it is omitted.

6 Points for Chris

1. $\mathbf{T} = \mathsf{R} + \mathsf{K}\,dt$, whereas $\mathbf{S} = \mathsf{J} + \mathsf{Q}\,dt$. The form parallel to S is $\mathsf{K} + \mathsf{R}\,dt$. With $dt = 1$ the two give the same number, but only then. Which is intended?
2. $dt = 1$ multiplies an acceleration that is then added to a velocity. The size of the anisotropy is set by the ratio $GM\,dt/(D^2kR)$, which is 0.247 here and depends on the time unit. What sets dt?
3. $G = 6.67 \times 10^{-17}$ is labelled N(km/kg)^2. In km and s, SI G is 6.67×10^{-20} km^3 kg^{-1} s^{-2}. Only the product GM enters, so this is equivalent to a 1000× larger M.
4. Migkas lookup: AG in row θ uses key $\theta + 10$ on `migkas!E`, a uniform grid of step 360/315 that indexes $\ell = 46 \ldots 360$. The model at θ is therefore compared with $\ell = 45 + \lfloor \frac{7}{8}(\theta + 10) \rfloor$, which is a 10° offset plus a 7/8 stretch; $\ell < 46°$ is never used. Is that intended, or should $\theta = \ell$? Against the Migkas points, R^2 is 0.63 with $\theta = \ell$ and 0.26 with the sheet mapping.
5. BA is a sine in θ with a set amplitude and phase, applied after the geometry. Is it meant as a physical term (rotation of the CoE axis), or as a fit factor?
6. Header "D" on column E is the CoE–P distance; the *terms* tab calls it "distance from earth to coe".
7. Column L equals kR in every row, so AF = U/(kR). Please confirm this normalisation is the intended A_{all}.
8. Which tab is current? The formulas are the same on all four; the parameters and R^2 differ (Table 1).

Files: `papers/spreadsheet_model/{columns.tex, recompute_columns.py, recompute_results.json, primary_columns.csv, make_figure.py, A_compare.pdf/png, chris_model.xlsx}`.